

A Discrete Metaplectic Obstruction for a Genus-Two Rauzy–Veech Transfer Proposal

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Abstract

We source-lock the labeled four-interval reversal permutation, reconstruct its seven-state hyperelliptic Rauzy diagram in the stratum $\mathcal{H}(2)$, and enumerate its primitive labeled directed cycles through elementary length twelve while preserving the chronological Kontsevich–Zorich cocycle. There are 828 primitive labeled free cycles and 146 phase-invariant eventually-positive labeled cycles. Twenty-one of the latter have $\det(I - M) = 0$, so the regular point formula for the Weil distribution character cannot provide a finite weight on the full selected labeled-cycle set. We do not claim that all codes are distinct primitive unmarked Teichmüller geodesics. We also prove that a nonzero absolutely norm-summable discrete sum of metaplectic atoms is noncompact after central-sign-aware aggregation. Together with an exact branch-compression lemma, this closes the unsmoothed ordinary Fredholm-determinant form within either of two explicit realization classes. The application hypothesis on a particular canonical analytic Zorich space remains open. The result does not rule out a genuinely distributional, semifinite, or continuously smoothed trace object.

1 Question and scope

The proposed operator combines a Rauzy–Veech–Zorich base transfer operator with the oscillator representation of its symplectic homology cocycle:

$$\mathcal{L}_s^{\text{Mp}} = \sum_h e^{-sr_h} K_h \otimes \mu(\tilde{B}_h). \quad (1)$$

Here h is a chronological return branch, r_h is its roof, and any metaplectic lift must retain its central sign. We ask only whether the *unsmoothed discrete operator* in (1) can be an ordinary nuclear Fredholm object. No prime or Riemann-zero data enter the construction.

The answer is negative in two natural and explicitly stated operator classes. The conclusion must not be enlarged to flat traces, distribution characters, semifinite determinants, or continuous group smoothing.

2 The literal Rauzy system

We fix

$$\pi_0 = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix}.$$

The prefix sets in its two rows disagree for every proper width, hence the permutation is irreducible. It is the four-letter central vertex of the hyperelliptic Rauzy diagram described in [1]; no permutation is substituted after inspection.

For a labeled permutation π , let Ω_π be the crossing matrix: an entry is $+1$ when the ordered pair is top-forward and bottom-reversed, -1 in the opposite case, and zero otherwise. At the base state,

$$\Omega_0 = \begin{pmatrix} 0 & 1 & 1 & 1 \\ -1 & 0 & 1 & 1 \\ -1 & -1 & 0 & 1 \\ -1 & -1 & -1 & 0 \end{pmatrix}, \quad \det \Omega_0 = 1.$$

Thus $\text{rank } \Omega_0 = 4 = 2g$. The relation $d = 2g + s - 1$ gives $s = 1$, and the unique zero has order $2g - 2 = 2$. Consequently the system belongs to $\mathcal{H}(2)$, with no relative-to-absolute kernel. These conventions follow [3, 4].

If an edge has winner w and loser ℓ , we use the forward homology matrix

$$B_e = I + E_{\ell,w}. \quad (2)$$

Later edges multiply on the left. An open edge satisfies

$$B_e \Omega_{\text{src}} B_e^T = \Omega_{\text{dst}}. \quad (3)$$

It generally does not preserve one fixed form. A closed loop at v instead satisfies

$$B_w^T J_v B_w = J_v, \quad J_v = \Omega_v^{-1}.$$

Our exact reconstruction has seven states and fourteen directed edges and checks (3) edge by edge.

3 Chronological periodic ledger

A cycle is encoded by directed edge tokens, quotiented by cyclic rotation but not by reversal. Proper powers are removed at the token level. Entrywise positivity is not invariant under a change of cyclic cut. We therefore call a free cycle eventually positive when every phase monodromy has a positive power. The four-dimensional Wielandt bound makes this an exact finite test; in the released range the largest observed exponent is three.

This selector also proves labeled Rauzy admissibility. If $M = B_{e_n} \cdots B_{e_1}$, the length-visitation product is $R = M^T$. For a positive Perron vector λ_0 of R , put $\lambda_n = \lambda_0 / \rho(R)$. The identity $\lambda_0 = R\lambda_n$, factored edge by edge, gives $\lambda_{i-1,w} = \lambda_{i,w} + \lambda_{i,\ell} > \lambda_{i,\ell} = \lambda_{i-1,\ell}$ at every declared winner. Thus the word returns projectively with the registered chronology, and the normalized roof telescopes to $\log(\|\lambda_0\|_1 / \|\lambda_n\|_1) = \log \rho(R)$.

Throughout, “primitive” means not a proper power as a closed directed edge-token word in the fixed-label Rauzy coding, modulo cyclic rotation only. We do not audit the possible identifications that can arise after forgetting labels, marked separatrices, or other marking data. The counts below are coded-orbit counts, not counts of pairwise distinct primitive closed Teichmüller geodesics in unmarked moduli space.

The independent producer and checker obtain the counts in Table 1. The checker uses both explicit cycle enumeration and the Möbius–trace identity

$$P_n = \frac{1}{n} \sum_{d|n} \mu(d) \text{tr}(A^{n/d})$$

for the seven-state adjacency matrix.

The unique selected length-eight cycle has characteristic polynomial

$$x^4 - 7x^3 + 13x^2 - 7x + 1$$

n	1	2	3	4	5	6	7	8	9	10	11	12
primitive	4	2	4	4	8	11	22	35	64	110	204	360
$M^q > 0$	0	0	0	0	0	0	0	1	6	14	36	89

Table 1: Exact primitive labeled directed free cycles and phase-invariant eventually-positive labeled cycles.

and

$$\det(I - M^r) = 1, 29, 361, 3509, 30976, 261725, \quad 1 \leq r \leq 6.$$

Its Perron root is isolated by exact rational endpoints near 4.3902568845156. The period candidate $\log \lambda_{\text{PF}}$ obeys exact repetition because $\rho(M^r) = \rho(M)^r$.

The 146 labeled periodic words realize only 41 reciprocal characteristic polynomials. We do not quotient by this homological spectral coincidence: primitive coded dynamics and matrix spectrum are different levels of information.

Proposition 3.1 (Fixed-vector singular cycles). *Among the 146 eventually-positive primitive labeled cycles through length twelve, exactly 21 satisfy $\det(I - M) = 0$: six at length ten, six at length eleven, and nine at length twelve. For each of them, $\det(I - M^r) = 0$ for every $r \geq 1$.*

Proof. The finite counts and factorizations are exact certificate statements checked by an independent implementation. Each listed reciprocal characteristic polynomial contains $(x - 1)^2$. Hence 1 remains an eigenvalue of every power, which proves the all-repetition claim. $\square$

4 Two noncompactness theorems

Theorem 4.1 (Tensor essential norm). *Let H_0, H_1 be Hilbert spaces, let $K : H_0 \rightarrow H_1$ be bounded, and let U be unitary on an infinite-dimensional Hilbert space F . Then*

$$\|K \otimes U\|_{\text{ess}} = \|K\|.$$

Proof. Fix a unit vector $x \in H_0$ and an orthonormal sequence (e_n) in F . Then $x \otimes e_n \rightarrow 0$. A compact operator C sends this sequence to zero in norm, while $\|(K \otimes U)(x \otimes e_n)\| = \|Kx\|$. Hence $\|K \otimes U - C\| \geq \|Kx\|$. Supremizing over x gives the lower bound; the reverse inequality is immediate. $\square$

Corollary 4.2 (Branch compression). *Under the assumptions of Theorem 4.1, let X, Y be Banach spaces and let $T : X \rightarrow Y$ be bounded. If bounded maps $J : H_0 \otimes F \rightarrow X$ and $R : Y \rightarrow H_1 \otimes F$ satisfy $RTJ = aK \otimes U + C$ with $a \in \mathbb{C}$, $aK \neq 0$, and C compact, then*

$$\|T\|_{\text{ess}} \geq \frac{|a|\|K\|}{\|R\|\|J\|} > 0.$$

The next result removes the need for cylinder projections on analytic spaces.

Theorem 4.3 (Discrete metaplectic atoms). *Let $d \in \mathbb{N}$, $d \geq 1$, let H_0, H_1 be Hilbert spaces, and let $\mu : \text{Mp}(2d, \mathbb{R}) \rightarrow \mathcal{U}(L^2(\mathbb{R}^d))$ be the oscillator representation. Let I be countable. For each $h \in I$, let $\tilde{g}_h \in \text{Mp}(2d, \mathbb{R})$ and let $A_h : H_0 \rightarrow H_1$ be bounded. Suppose*

$$T = \sum_{h \in I} A_h \otimes \mu(\tilde{g}_h), \quad \sum_{h \in I} \|A_h\| < \infty.$$

Let $\pi : \text{Mp}(2d, \mathbb{R}) \rightarrow \text{Sp}(2d, \mathbb{R})$ be the projection. Choose one lift $\tilde{g}$ above every projected atom g and write

$$\mu(\tilde{g}_h) = \varepsilon_h \mu(\tilde{g}), \quad A_g = \sum_{\pi(\tilde{g}_h)=g} \varepsilon_h A_h, \quad \varepsilon_h \in \{\pm 1\}.$$

If some $A_g \neq 0$, then T is noncompact. More precisely,

$$\|T\|_{\text{ess}} \geq \sup_{\|x\|=\|y\|=1} \left(\sum_g |\langle A_g x, y \rangle|^2 \right)^{1/2}.$$

Proof. Fix unit vectors $x \in H_0$, $y \in H_1$ and slice the base variables. It remains to study $V = \sum_g a_g \mu(\tilde{g})$ with $a_g := \langle A_g x, y \rangle$ and $(a_g) \in \ell^1$. Choose a generic phase-space vector z outside the countable union of the proper subspaces $\ker(g - h)$ for distinct atoms. For a unit Schwartz vector ϕ , put $\phi_t = W(tz)\phi$, where W is the Weyl representation. Then $\phi_t \rightharpoonup 0$.

Metaplectic covariance gives

$$\mu(\tilde{g})W(tz)\phi = W(tgz)\mu(\tilde{g})\phi$$

up to convention-dependent phases. Weyl matrix coefficients vanish at infinity, so off-diagonal inner products tend to zero because $t(g - h)z \rightarrow \infty$. The ℓ^1 hypothesis dominates the double series and yields

$$\|V\phi_t\|^2 \longrightarrow \sum_g |a_g|^2.$$

Compact operators send ϕ_t to zero in norm. This proves the scalar essential-norm bound; the input and output slices have norm one. $\square$

5 Distribution character versus ordinary trace

For a unitary on an infinite-dimensional space, $|U| = I$ and $\text{Tr } |U| = \infty$. Thomas [2] defines the Weil character distribution after smooth integration over the metaplectic group. On the regular set $\det(g - I) \neq 0$, that distribution has a smooth representative whose amplitude is $|\det(g - I)|^{-1/2}$ and whose phase retains the lift and Weil/Maslov data. It is not the ordinary trace of the isolated unitary.

Proposition 3.1 shows that the regular point formula cannot be used on the entire selected labeled-cycle set. Those 21 cycles are not to be deleted or assigned a finite value by a cutoff. Any surviving construction must specify a genuine distributional or generalized trace operation.

6 Decision

Corollary 4.2 closes every bounded realization of (1) that admits the stated nonzero branch compression, either exactly or modulo compact operators. Theorem 4.3 also closes any Hilbert-space analytic realization on a half-plane where the branch norms are absolutely summable and one signed aggregate is nonzero. These hypotheses have not yet been verified on a particular canonical analytic Zorich space. Within either stated class, noncompactness excludes all finite Schatten classes and ordinary or finite-order regularized Fredholm determinants for this operator.

This is a scoped obstruction, not a Hilbert–Pólya construction. It does not assert that $I - z\mathcal{L}_s^{\text{Mp}}$ is never Fredholm or invertible, and it does not exclude flat traces, distribution characters, canonical semifinite determinants, continuous group smoothing, or a different quantum fiber. The Route-A verdict is

$$(A1_WEAK, A2_FAIL, A3_FAIL, A4_FORMAL_HINT),$$

so Route B is not opened.

Reproducibility

The producer emits the complete exact certificate. A checker that does not import the producer rebuilds the seven-word automaton and verifies the cycle counts by Möbius inversion. Exact integers and rational root intervals are authoritative; displayed decimals are diagnostics. The accompanying mutation suite rejects transposed edge matrices, reversed chronology, move-word-only cycle identities, proper powers, phase-dependent filtering, and finite weighting of the singular character locus.

References

- [1] Artur Avila, Carlos Matheus, and Jean-Christophe Yoccoz. Zorich conjecture for hyperelliptic rauchy–veech groups. *Mathematische Annalen*, 370:785–809, 2018.
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- [4] Anton Zorich. Flat surfaces. In *Frontiers in Number Theory, Physics, and Geometry I*, pages 437–583. Springer, 2006.